THE SHAPE–HEIGHT TRANSFORM: A GEOMETRICALLY INJECTIVE FRAMEWORK FOR FUNCTION RECOVERY FROM HORIZONTAL CROSS-SECTIONS

Abstract

We introduce the Shape–Height Transform (SHT), a geometric functional that assigns to a real-valued function f the collection of area-and-moment measurements of the planar regions enclosed between the graph of f and horizontal slicing lines at varying heights. For strictly convex bowl-shaped functions, we prove that SHT is fully injective: equality of SHT implies equality of the underlying functions. We then extend the construction to the general class of piecewise-convex functions with finitely many valleys, introducing the Multi-Interval SHT (SHT_*), and prove a comprehensive injectivity theorem:

SHT_* uniquely identifies any function in this class up to a trivial vertical shift. These results demonstrate that the geometry of horizontal slices contains sufficient information to recover the entire analytic structure of a function. Consequently, SHT yields a robust general alternative to scalar geometric measures such as GMR (Geometric Mean Radius), eliminating the loss of information inherent in single-number descriptors and enabling complete recovery of roots, tangents, curvature, and higher-order behaviour from geometric cross-sections alone.

1 Introduction

The geometry of the region between a function and an enclosing curve is a classical object of study in analysis. If one intersects a graph $y = f(x)$ with a horizontal line $y = h$, the resulting shape consists of one or more planar "caps" whose width and distribution encode information about the inverse branches of f. Classical geometric invariants such as area, perimeter, centroid, or the Geometric Mean Radius extract only partial information from such shapes. These scalars are not injective: different functions can generate shapes with identical values of these invariants.

This paper introduces a geometric functional that does not lose information.

We define the Shape–Height Transform (SHT) as a family of measurements indexed by height h: for each h, we compute (i) the area of the region

between the graph and the line $y = h$ and (ii) the first horizontal moment of that region. These functions of height form a geometric signature of the entire curve. We prove that, for convex bowls, SHT uniquely determines the original function. We then extend the transform to functions with finitely many valleys by using height-indexed vectors of areas and moments corresponding to multiple disjoint caps. We prove an injectivity result for the general class.

These results show:

> Horizontal slicing contains enough geometric information to reconstruct the entire function.

The implications are strong: every strictly convex or finitely-valley function can be recovered from purely geometric data, without solving differential or integral equations and without spectral analysis.

2 Preliminaries

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. For a fixed height h, define the horizontal level set

$$L(f, h) := \{x \in \mathbb{R} : f(x) = h\}.$$

Assume for each h the set $L(f, h)$ consists of finitely many points, ordered as

$$x_1(h) < x_2(h) < x_3(h) < \cdots < x_{2n(h)}(h).$$

Between these intersections, the region bounded by the graph and the line $y = h$ is composed of disjoint pieces

$$\Omega_f^{(j)}(h) = \{(x, y) : x_{2j-1}(h) \leq x \leq x_{2j}(h),\ f(x) \leq y \leq h\}, \quad j = 1, \ldots, n(h).$$

Every $\Omega_f^{(j)}(h)$ is a planar region shaped like an inverted bowl truncated at height h. Our transform will use geometric quantities of the $\Omega_f^{(j)}(h)$.

3 The single-valley case

3.1 Function class

Let $\mathcal{C}_1$ be the set of C^1 functions $f : \mathbb{R} \to \mathbb{R}$ satisfying:

1. $f(x) \to \infty$ as $|x| \to \infty$,
2. f has a unique global minimum at x_0,
3. f is strictly convex.

Then for all heights $h > f(x_0)$, there are exactly two intersection points of $y = f(x)$ with the line $y = h$:

$$a_f(h) < x_0 < b_f(h), \qquad \text{with } f(a_f(h)) = f(b_f(h)) = h.$$

Define the single region:

$$\Omega_f(h) = \Omega_f^{(1)}(h) = \{(x, y) : f(x) \leq y \leq h,\ a_f(h) \leq x \leq b_f(h)\}.$$

3.2 Area and moment

Define the geometric invariants:

$$A_f(h) = \int_{a_f(h)}^{b_f(h)} (h - f(x))\, dx,$$

$$M_f(h) = \int_{a_f(h)}^{b_f(h)} x\,(h - f(x))\, dx.$$

3.3 Definition (SHT in the single-valley case)

$$\boxed{\mathcal{S}(f) := (A_f(h), M_f(h))_{h \geq f(x_0)}}$$

3.4 Recovering level-set endpoints from SHT

Differentiation under the integral gives:

Lemma 1. *For all $h > f(x_0)$,*

$$\frac{d}{dh} A_f(h) = b_f(h) - a_f(h),$$

$$\frac{d}{dh} M_f(h) = \frac{b_f(h)^2 - a_f(h)^2}{2}.$$

Define:

$$W_f(h) = b_f(h) - a_f(h), \quad S_f(h) = b_f(h)^2 - a_f(h)^2.$$

Then

$$W_f(h) = A'_f(h), \quad C_f(h) = \frac{M'_f(h)}{A'_f(h)},$$

and

$$a_f(h) = C_f(h) - \tfrac{1}{2}W_f(h), \qquad b_f(h) = C_f(h) + \tfrac{1}{2}W_f(h).$$

Thus SHT uniquely determines $a_f(h)$ and $b_f(h)$ for all h.

3.5 Main theorem (injectivity, single-valley)

Theorem 1. *Let $f, g \in \mathcal{C}_1$. Suppose that after aligning minima vertically,*

$$\mathcal{S}(f) = \mathcal{S}(g).$$

Then

$$f(x) = g(x) \quad \textit{for all } x \in \mathbb{R}.$$

Proof sketch. Equality of A_f and A_g and of M_f and M_g implies equality of their derivatives, which implies equality of widths and centers at all heights, hence equality of $a_f(h), b_f(h)$ and $a_g(h), b_g(h)$. These functions are exactly the inverse branches of f and g, so equality of the branches implies equality of the full functions. □

Thus SHT is injective on strictly convex bowls.

4 Generalisation to piecewise-convex functions (multiple valleys)

4.1 Class of functions

Let $\mathcal{C}_{\text{pw}}$ be the set of C^1 functions with finitely many disjoint valleys:

1. $f(x) \to \infty$ as $|x| \to \infty$,
2. There exist intervals $I_1, \ldots, I_m$ such that f is strictly convex on each I_k, with a single local minimum within each I_k,
3. Between consecutive valleys the function has local maxima.

For each height $h > \min f$, the horizontal line $y = h$ intersects the graph in $2n(h)$ points:

$$x_1(h) < x_2(h) < \cdots < x_{2n(h)}(h).$$

4.2 Multi-interval areas and moments

For $j = 1, \ldots, n(h)$, define:

$$A_f^{(j)}(h) = \int_{x_{2j-1}(h)}^{x_{2j}(h)} (h - f(x))\, dx,$$

$$M_f^{(j)}(h) = \int_{x_{2j-1}(h)}^{x_{2j}(h)} x\,(h - f(x))\,dx.$$

As height increases, components may split or merge, but the ordering of caps by left endpoint is always well-defined.

4.3 Definition (multi-interval SHT)

$$\boxed{\mathcal{S}_*(f) := \{(A_f^{(1)}(h), M_f^{(1)}(h), \ldots, A_f^{(n(h))}(h), M_f^{(n(h))}(h)) : h \geq \min f\}}$$

4.4 Reconstruction

The differentiation formulas remain valid componentwise:

$$\frac{d}{dh} A_f^{(j)}(h) = x_{2j}(h) - x_{2j-1}(h),$$

$$\frac{d}{dh} M_f^{(j)}(h) = \frac{x_{2j}(h)^2 - x_{2j-1}(h)^2}{2}.$$

Consequently, each interval's width and center are recovered, hence all endpoints $x_i(h)$ for all h, including at merging heights.

Thus SHT_* reconstructs all level-set endpoints, and therefore reconstructs all inverse branches of f.

4.5 Main theorem (injectivity, multiple valleys)

Theorem 2. *Let $f, g \in \mathcal{C}_{pw}$. After aligning minima vertically, if*

$$\mathcal{S}_*(f) = \mathcal{S}_*(g),$$

then

$$f(x) = g(x) \quad \textit{for all } x \in \mathbb{R}.$$

Proof sketch. Equality of SHT_* gives equality of all $A_f^{(j)}$ and $M_f^{(j)}$, hence equality of widths and centers of each cap, hence equality of endpoints of every level-set interval at all heights. Level-set endpoints are inverse branches of the function, so equality of all inverse branches implies equality of f and g everywhere. □

Therefore SHT_* is a complete geometric fingerprint for any piecewise-convex function with finitely many valleys.

5 Consequences

5.1 Root recovery

Roots of f are simply the endpoints of the level set at $h = 0$, obtained from reconstructed $a_f(0), b_f(0)$.

5.2 Tangent recovery

The tangent at any location x is obtained by differentiating f; since f itself is reconstructed, all tangents are recovered exactly.

5.3 Curvature and higher-order behaviour

Once f and f' are determined, curvature κ, third derivative, etc., follow immediately.

5.4 Numerical reconstruction

The transform is inversion-friendly:

- A_f and M_f require simple integration,
- inversion uses differentiation and algebraic formulas only (no PDEs or eigenvalue problems).

6 Conclusion

The Shape–Height Transform (SHT) is a general and fully injective geometric transform for recovering real functions from planar geometry. While scalar geometric descriptors such as the Geometric Mean Radius inevitably lose structural information, SHT and its multi-interval extension capture the full level-set geometry, enabling explicit reconstruction of roots, tangents, curvature, and higher-order analytic behaviour.

This establishes a new principle:

> The family of shapes formed by horizontal slicing of a function determines the function completely.

Future directions include:

- numerical schemes for high-accuracy inversion,
- extension to functions in higher dimensions using volume- and moment-height transforms,
- noise-robust reconstruction algorithms for experimental shape data.